THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE UNIT TEST SET 1

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A toy is a cone of base radius 7 cm and height 24 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 858 cm² **(b)** 1012 cm² **(c)** 550 cm² **(d)** 836 cm²

2. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70 with frequencies 3, 16, 20, 4, 7 respectively, The mean of the data is: 1

(a) 442 **(b)** 44.2 **(c)** 45 **(d)** 45.2

3. Two dice are thrown simultaneously. The probability that the sum of the numbers is 8 is: 1

(a) $\frac{5}{36}$ **(b)** $\frac{1}{9}$ **(c)** $\frac{31}{36}$ **(d)** $\frac{1}{6}$

4. A capsule is a cylinder of radius 3.5 cm and length 10 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 748 cm² **(b)** 451 cm² **(c)** 297 cm² **(d)** 374 cm²

5. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 with frequencies 4, 3, 2, 4, 8, 4 respectively, The median of the data is: 1

(a) 73.4 **(b)** 80 **(c)** 70 **(d)** 78.75

6. A coin is tossed three times. The probability of getting exactly two heads is: 1

(a) $\frac{3}{8}$ **(b)** $\frac{5}{8}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{1}{2}$

7. A solid is a cube of edge 14 cm surmounted by a hemisphere of diameter 14 cm (the hemisphere rests on a face of the cube). The total surface area of the solid is: 1

(a) 1176 cm² **(b)** 1484 cm² **(c)** 1022 cm² **(d)** 1330 cm²

8. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90 with frequencies 9, 4, 10, 13, 4 respectively, The mode of the data is: 1

(a) 64.75 **(b)** 70 **(c)** 72.5 **(d)** 80

9. A letter is chosen at random from the word 'MATHEMATICS'. The probability that it is a vowel is: 1

(a) $\frac{3}{11}$ **(b)** $\frac{7}{11}$ **(c)** $\frac{4}{11}$ **(d)** $\frac{5}{11}$

10. From a solid cylinder of height 24 cm and base radius 7 cm, a conical cavity of the same height and the same base radius is hollowed out. The total surface area of the remaining solid is: 1

(a) 1606 cm² **(b)** 1364 cm² **(c)** 1760 cm² **(d)** 1914 cm²

11. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 3, 8, 20, 12, 7 respectively, The modal class is: 1

(a) 10–20 **(b)** 40–50 **(c)** 20–30 **(d)** 30–40

12. A bag contains 4 red and 6 blue balls. 2 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{7}{12}$ **(b)** $\frac{5}{12}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{6}{13}$

13. A wooden article is made by scooping out a hemisphere of radius 3.5 cm from each end of a solid cylinder of height 10 cm and base radius 3.5 cm. The total surface area of the article is: 1

(a) 297 cm² **(b)** 220 cm² **(c)** 374 cm² **(d)** 451 cm²

14. The mean of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 3, p, 20, 4, 7 respectively is 44.2. The value of p is: 1

(a) 18 **(b)** 21 **(c)** 15 **(d)** 16

15. A box contains 20 tickets numbered 1 to 20. One ticket is drawn at random. The probability that the number drawn is a multiple of 3 is: 1

(a) $\frac{7}{20}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{3}{10}$ **(d)** $\frac{7}{10}$

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A solid metallic sphere of radius 6 cm is melted and drawn into a wire of circular cross-section of radius 2 cm. Find the length of the wire. 2

17. The marks obtained by students in a test are given below. 2

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	20	22	20	17	6	15

Find the mean.

18. A coin is tossed three times. Find the probability of getting at least one tail. 2

19. A solid metal cuboid 24 cm by 12 cm by 9 cm is melted and recast into cubes of edge 6 cm. Find the number of cubes obtained. 2

20. The weights (in kg) of students of a class are given below. 2

Weight (kg)	70–80	80–90	90–100	100–110	110–120
Number of students	13	16	7	7	7

Find the mode.

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A toy is a cone of base radius 21 cm and height 10 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

22. The marks obtained by students in a test are given below. 3

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	20	22	20	17	6	15

Find the mean.

23. Two dice are thrown simultaneously. Find the probability that the product is a perfect square. 3

24. A capsule is a cylinder of radius 10.5 cm and length 10 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

OR

The ages (in years) of patients admitted to a hospital in a month are given below. 3

Age (years)	0–10	10–20	20–30	30–40	40–50
Number of patients	13	15	9	8	5

Find the median.

25. A box contains 18 tickets numbered 1 to 18. One ticket is drawn at random. Find the probability that the number drawn is a multiple of 3. 3

OR

From a solid cylinder of height 12 cm and base radius 7 cm, a conical cavity of the same height and base radius is hollowed out. Find the volume of the remaining solid. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Water from a full cylindrical tank of radius 7 cm and height 10 cm is poured into an empty rectangular tank of base 11 cm by 7 cm. Find the height of water in the rectangular tank. 5

OR

The mean of a distribution with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 and frequencies 4, 3, 2, p, 8, 4 respectively is 73.4. Find the value of p. 5

27. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the sum of the numbers is 8,
(ii) both dice show the same number (a doublet),
(iii) the product of the numbers is 12.

OR

A conical vessel of radius 6 cm and height 8 cm is full of water. The water is poured into a cylindrical vessel of base radius 4 cm. Find the height of the water in the cylindrical vessel. 5

END OF QUESTION PAPER — Total: 50 Marks

MASTERMIND